Predicting coverage-standardized plant diversity from raw Landsat phenology curves across a unified Chilean vegetation plot network

1 Methods

1.1 Study data and diversity facets

We combined two vegetation plot networks spanning central and southern Chile. Parcelas-CL (Cerda-Paredes et al., 2026) is a community-sourced database of 1082 plots between 30°S and 38°S. Living Trees is a forest inventory of 2020 plots with a substantially wider geographic extent. We merged both networks into a unified pool of 3102 plots, matching plot identifiers across networks by exact coordinates.

We derived three diversity facets following the methodology of Pérez-Giraldo et al. (2026), applied to the subset of the unified pool with genuine individual-level abundance counts (479 Parcelas-CL plots recorded under the Abundance stratum, plus individual tree counts from Living Trees). We estimated the local contribution to beta diversity (LCBD) using the quantitative Sørensen decomposition of Podani and Schmera, implemented in the *adespatial* package (`beta.div.comp(coef="S", quant=TRUE)` followed by `LCBD.comp(sqrt.D=TRUE)`; Legendre and De Cáceres, 2013). This facet (*lcbd_count_sorensen*) covers 2499 of the 3102 plots. We estimated coverage-standardized phylogenetic and taxonomic diversity with iNEXT.3D (Chao et al., 2021), following Pérez-Giraldo et al. (2026): `estimate3D(base="coverage", level=NULL, nboot=1)` for Hill orders $q = 0, 1, 2$. Phylogenetic diversity (*pd_inext_q0/q1/q2*) additionally requires restricting the species pool to a 610-tip phylogeny built with V.PhyloMaker2 (Jin and Qian, 2022) and covers 888 plots; taxonomic diversity (*td_inext_q0/q1/q2*), without that restriction, covers 895. The coverage-based estimator requires a minimum of five observed species per sampling unit, which accounts for the lower coverage of both facets relative to *lcbd_count_sorensen*. We reindexed all three facets onto the full 3102-plot pool, leaving missing values where the estimator did not meet its species floor, so that no plot row is ever dropped.

1.2 Remote sensing predictors and topography

We extracted Landsat surface reflectance time series (NDVI, EVI, kNDVI, NBR and SAVI) for the centre pixel of a 5×5 Landsat window at each plot. We interpolated each raw series onto a regular grid of 100 steps with a five-observation moving-average smoothing window, without fitting any parametric composite curve; a preliminary comparison against a composite phenological curve showed the raw interpolated curve consistently outperforming the composite, so we used the raw curve throughout.

In addition to the phenological curve, we built a context block from digital-elevation-model-derived topography (elevation, slope, northness, eastness, heat-load index, topographic position index, ruggedness and curvature, each summarised as the window mean and standard deviation) and plot area. We extracted Living Trees topography independently and merged it with the Parcelas-CL topography through an exact coordinate match, removing a previous median-imputation step that had affected all 2020 Living Trees plots.

1.3 Cross-validation

We evaluated every model with `kfold5_block20_unified`: five folds grouped by 20×20 km geographic blocks over a Chile-centred equal-area projection, following the spatial-blocking principle for avoiding leakage from spatial autocorrelation (Roberts et al., 2017). This blocking scheme groups the 3102 plots into 610 distinct blocks.

1.4 Predictive models

We compared four model families under the same design (raw kNDVI curve, centre pixel, topography+area context, `kfold5_block20_unified`):

- Random Forest (Breiman, 2001) on the concatenated curve and context.
- A 1-D convolutional network over the 100-step curve.
- A 2-D convolutional network (separable architecture, ∼15k parameters) over the curve reshaped as an image via a serpentine transform.
- The same 2-D network, initialised from a Masked Autoencoder (He et al., 2022) pretrained on kNDVI curves from the non-unified pool.

We trained each model with three to five random seeds and report the pooled out-of-sample R^2 across plots.

2 Results

2.1 Vegetation index selection

We swept all five vegetation indices under the same design (raw curve, centre pixel, real topography, `kfold5_block20_unified`). We averaged R^2 only over the three facets with usable signal ($R^2 > 0.4$: *lcbd_count_sorensen*, *pd_inext_q0*, *td_inext_q0*); the remaining facets show no signal under any index, and averaging them with the informative facets would have masked the only signal present.

Index	*lcbd_count_sorensen*	*pd_inext_q0*	*td_inext_q0*	Mean ($R^2 > 0.4$)
kNDVI	0.434	0.608	0.772	**0.6047**
EVI	0.426	0.612	0.775	0.6043
NDVI	0.433	0.608	0.767	0.6027
SAVI	0.424	0.615	0.763	0.6007
NBR	0.435	0.604	0.725	0.5880

Table 1: R^2 by vegetation index (raw curve, centre pixel, `kfold5_block20_unified`).

kNDVI achieved the highest mean R^2, by a narrow margin over EVI (0.6047 versus 0.6043); NBR ranked lowest, driven by its weaker performance on *td_inext_q0*. We adopted kNDVI as the default index for all subsequent analyses.

2.2 Effect of real Living Trees topography

Replacing the median-imputed topography of Living Trees plots with measured topography increased *lcbd_count_sorensen* by 0.02–0.04 across all five indices, with no notable effect on *pd_inext_q0* or *td_inext_q0*. We retained real topography as the default configuration.

2.3 Model comparison

With kNDVI fixed, we compared the four model families under `kfold5_block20_unified`:

Facet	2D-CNN	RF	1D-CNN	2D-CNN + MAE
lcbd_count_sorensen	0.434	**0.447**	0.420	0.437
pd_inext_q0	0.608	0.582	0.612	**0.613**
pd_inext_q1	0.011	0.004	**0.049**	-0.010
pd_inext_q2	0.065	0.054	**0.091**	0.040
td_inext_q0	0.772	0.666	0.780	**0.786**
td_inext_q1	-0.178	**0.042**	-0.144	-0.181
td_inext_q2	-0.079	-0.073	-0.066	-0.090

Table 2: Model family comparison (R^2, kNDVI, raw curve, centre pixel, `kfold5_block20_unified`).

The masked-autoencoder-initialised network achieved the highest absolute R^2 on *pd_inext_q0* (0.613) and *td_inext_q0* (0.786), without transferring that gain to *lcbd_count_sorensen* or to $q1/q2$. The 1-D network matched or exceeded the 2-D network on richness despite having fewer parameters (14,535 versus 15,175) and was the only family with usable signal on *pd_inext_q1/q2* (0.049 and 0.091). Random Forest trained in 114 s against 8–10 min for each neural network and won on *lcbd_count_sorensen* (0.447), though it lost phylogenetic and taxonomic richness to all three neural architectures.

2.4 Synthesis

Coverage-standardized richness (*pd_inext_q0*, *td_inext_q0*) is the only facet family robustly predictable under this design, with R^2 between 0.58 and 0.79 across all four model families and all five indices tested. *lcbd_count_sorensen* offers moderate, stable signal (0.39–0.45). Abundance- and dominance-weighted facets (*pd_inext_q1/q2*, *td_inext_q1/q2*) show no usable signal under any model, index or architecture tested – a limitation of the signal available in those specific facets, not of model architecture.

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